University of East London

Dissertation Proposal

Name of Student		
Department		
Registration No.		Date
Name of (i) Research Supervisor		
Title of Research Proposal	**AI-Powered Detection and Mitigation of Cross-Site Scripting (XSS) Attacks**	

Project Summary

Cross-Site Scripting (XSS) is one of the most common web vulnerabilities that attackers commonly exploit by injecting malicious scripts into some otherwise trusted websites. When these scripts are run within the browser of a user it can either result in information curtailing, session kidnapping or even page defacements. The scope of this project is to design and develop an AI-based detection and mitigation system with the help of supervised machine learning and natural language processing technique (Alsaffar et al., 2022). Inputs to the web will be analyzed by the system and script that can cause harm would be flagged, or neutralized in real-time. Here we will apply labeled dataset of XSS in which properties like character pattern and script behaviors will help us to train and evaluate model including Random Forest, SVM and LSTM (Hou et al., 2023). The model will be carried out in Python on Jupyter Notebook and will be measured with traditional indicators. Finally, the solution is supposed to enhance proactive web applications security and suggest a contextually smart mitigation mechanism appropriate to suit the present-day dynamic environments.

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